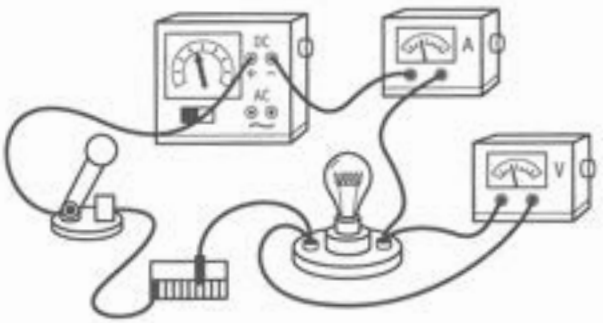
Circuit symbols 4a

An easy way to show how electrical devices are connected to a power source is by drawing a circuit diagram. A circuit diagram is made up of circuit symbols. These symbols are common and accepted by people from countries all over the world. The components or parts that make up a circuit can be classified into four groups:

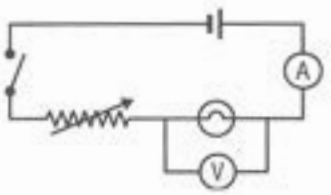
1. **Energy suppliers**: these supply the electrical energy to make the circuit work.
2. **Measuring devices**: these components make measurements in a circuit.
3. **Energy users**: these devises change electrical energy into other forms of energy, such as heat or light.
4. **Circuit controllers**: these can turn off the current or control the amount of energy in a circuit.

1 Carefully observe the two diagrams A and B below.

A



B



a What is similar about them? _____

b What is different about them?

_____

2 Use diagrams A and B above to work out the symbols of the electrical devices pictured. Draw the symbols in the space provided in the table.

Part	Use	Symbol	Group
Ammeter	Measures electrical current		
Battery or powerpack	Produces electrical energy		
Globe	Changes electrical energy into light and heat		
Switch	Is a gate in the circuit used by people to turn off the current		
Variable resistor	Controls the amount of energy in the circuit		
Voltmeter	Measures electrical potential difference		

3 Into which of the four circuit groups could each device be classified? Add your answers to the last column of the table.

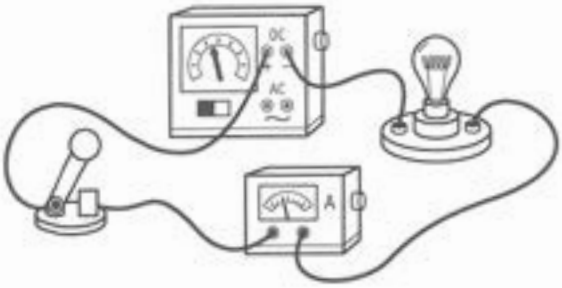
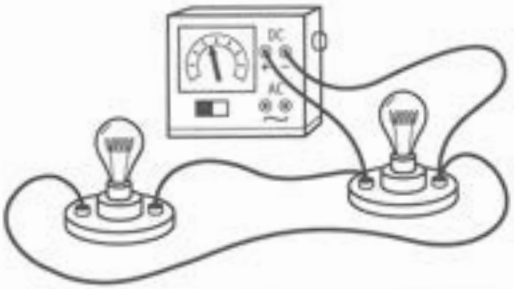
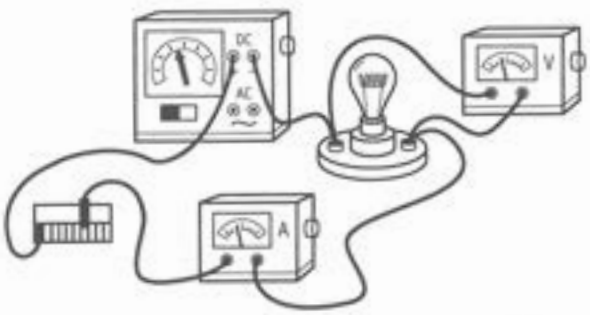
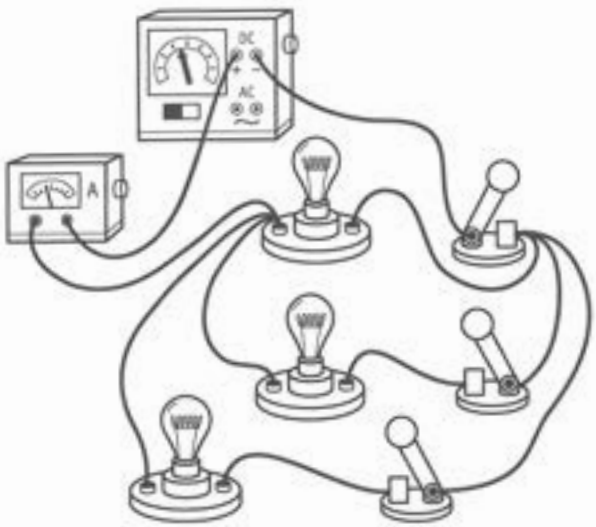
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Circuit symbols

4b

4 Why do scientists or technicians prefer to use the style of diagram B?

5 Draw a circuit diagram for the following set-ups.

Circuit	Circuit diagram
	
	
	
	

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